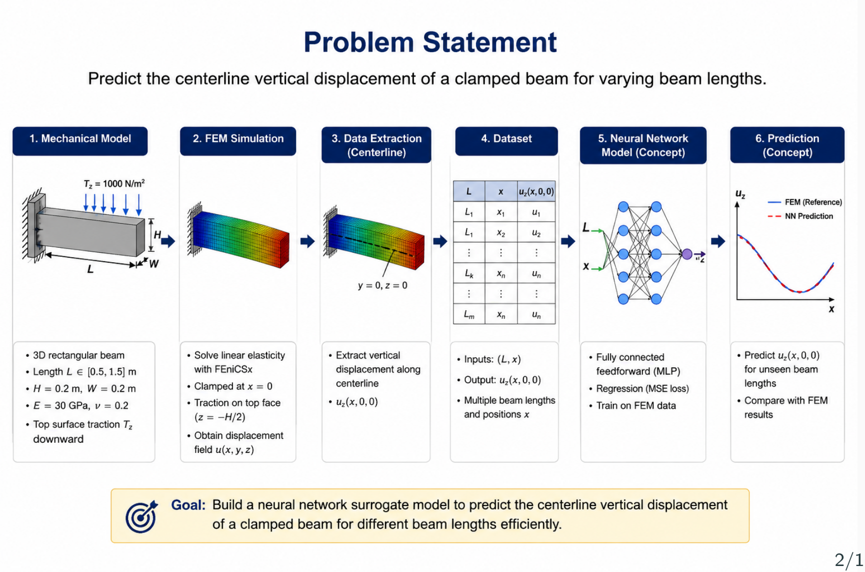
Parametric FEM and Neural Network Approximation

BAM Germany – Interview Presentation

Muhammad Adnan Anwar

June 2, 2026

From high-fidelity FEM simulations to fast predictive models



Problem Statement
Predict the centerline vertical displacement of a clamped beam for varying beam lengths.
1. Mechanical Model
T_z = 1000 N/m²
H
L
W
2. FEM Simulation
3. Data Extraction (Centerline)
$y = 0, z = 0$
4. Dataset
L | x | $u_z(x,0,0)$
L_1 | x_1 | u_1
L_1 | x_2 | u_2
L_k | x_n | u_n
L_m | x_n | u_n
5. Neural Network Model (Concept)
L
x
6. Prediction (Concept)
u_z
FEM (Reference)
NN Prediction
x
• 3D rectangular beam
• Length $L \in [0.5, 1.5]$ m
• $H = 0.2$ m, $W = 0.2$ m
• $E = 30$ GPa, $\nu = 0.2$
• Top surface traction T_z downward
• Solve linear elasticity with FEniCSx
• Clamped at $x = 0$
• Traction on top face ($z = -H/2$)
• Obtain displacement field $u(x, y, z)$
• Extract vertical displacement along centerline
• $u_z(x, 0, 0)$
• Inputs: (L, x)
• Output: $u_z(x, 0, 0)$
• Multiple beam lengths and positions x
• Fully connected feedforward (MLP)
• Regression (MSE loss)
• Train on FEM data
• Predict $u_z(x, 0, 0)$ for unseen beam lengths
• Compare with FEM results
Goal: Build a neural network surrogate model to predict the centerline vertical displacement of a clamped beam for different beam lengths efficiently.

Why a Surrogate Model?

Finite Element Method (FEM)

- High accuracy for structural mechanics
- Requires mesh generation and numerical solver
- Computationally expensive for repeated evaluations
- Not suitable for real-time prediction

Limitation

- Parametric studies require many FEM solves
- Design optimization becomes slow

Surrogate Model (Neural Network)

- Learns mapping from FEM data
- Approximates: $(L, x) \rightarrow u_z$
- Orders-of-magnitude faster inference
- Enables real-time prediction

Key Advantage

- Fast evaluation after training
- Enables optimization loops and design exploration
- Preserves FEM-driven physical trends

Goal: Replace repeated FEM simulations with a fast data-driven approximation.

Problem Formulation

Geometry

3D beam domain with variable length

$$\Omega = [0, L] \times \left[-\frac{W}{2}, \frac{W}{2}\right] \times \left[-\frac{H}{2}, \frac{H}{2}\right]$$

Governing Equations

equilibrium / force balance

$$-\nabla \cdot \sigma(u) = 0$$

constitutive law / Hooke's law)

$$\sigma(u) = \lambda \operatorname{tr}(\varepsilon(u)) I + 2\mu\, \varepsilon(u)$$

strain definition

$$\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T)$$

Material Parameters

E : Young's modulus (stiffness)

ν : Poisson ratio

$$E = 30\,\text{GPa}, \quad \nu = 0.2$$

$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

Boundary Conditions

$$u = 0 \quad \text{at } x = 0$$

$$\sigma(u)n = (0, 0, T_z) \quad \text{at } z = -\frac{H}{2}$$

Quantity of Interest

$$u_z(x, 0, 0)$$

Weak Formulation and FEM Discretization

Weak Formulation

Find $u \in V$ such that:

$$a(u, v) = L(v) \quad \forall v \in V_0$$

where

$$V = \{u \in [H^1(\Omega)]^3 \ : \ u = 0 \text{ on } \Gamma_D\}$$

$$V_0 = \{v \in [H^1(\Omega)]^3 \ : \ v = 0 \text{ on } \Gamma_D\}$$

$$a(u, v) = \int_\Omega \sigma(u) : \varepsilon(v) \, dx$$

$$L(v) = \int_{\Gamma_t} T \cdot v \, ds$$

Finite Element Discretization

- Implemented using **FEniCSx / DOLFINx**
- Vector-valued P_1 (linear) Lagrange elements
- Hexahedral mesh discretization of Ω
- Essential BC imposed on $\Gamma_D = \{x = 0\}$

Mesh Strategy and Dataset Generation

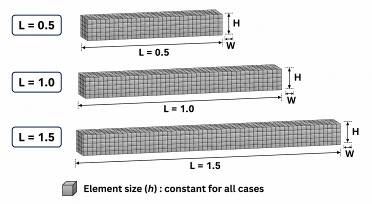
Mesh Scaling Strategy

Reference mesh determined via convergence study.

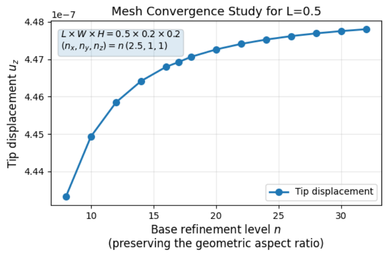
$$n_x(L) = n_{x,\mathrm{ref}} \frac{L}{L_{\mathrm{ref}}}, \qquad n_y, n_z = \mathrm{const.}$$

Maintains approximately uniform element size and aspect ratio for all beam lengths.

Mesh Strategy



Reference Mesh Convergence



Convergence-Based Mesh

- Mesh density selected from convergence study
- Ensures numerical accuracy
- Avoids over-refinement
- Used consistently across all L

FEM Validation

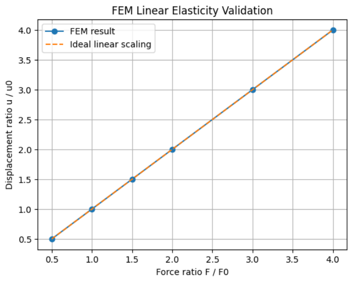
Linearity Check

- Verified proportionality between load and displacement
- Confirms linear elastic response of FEM model
- Ensures physical consistency of simulation

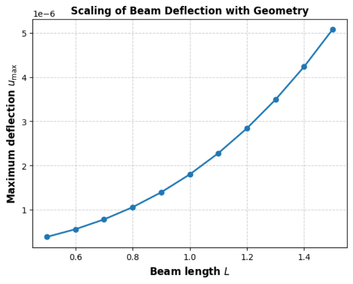
Geometry Response

- Maximum deflection evaluated across beam lengths
- Clear dependence of deformation on geometry
- Consistent trends across parameter space

Linearity Validation

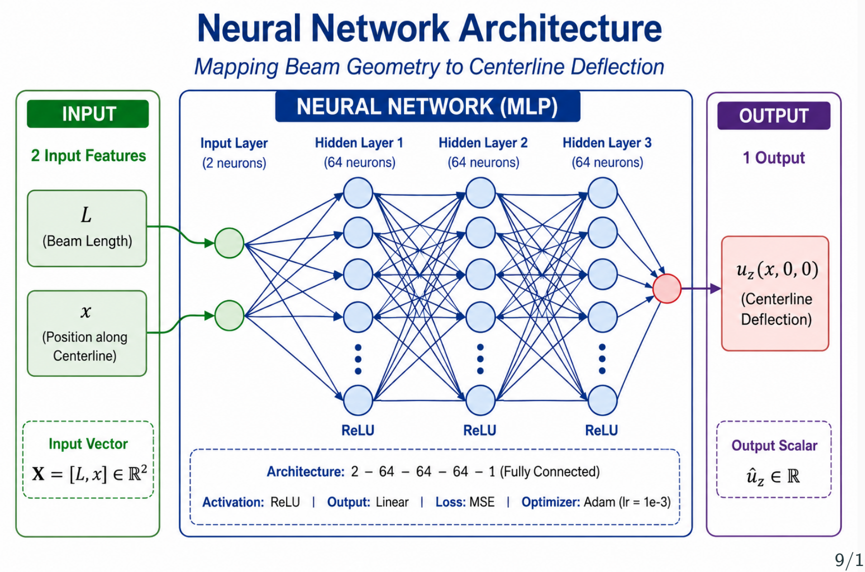


Deflection vs Geometry



Beam Animation

Neural Network

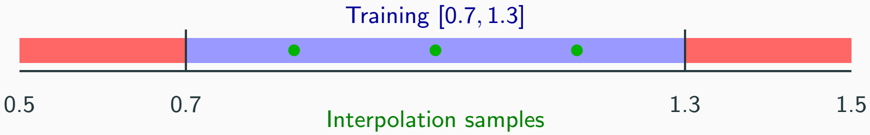


Training and Evaluation Strategy

Strategy

The model is trained on a central subset of beam geometries and evaluated under both interpolation and extrapolation regimes.

$$L \in [0.5,\, 1.5] \;=\; [0.5,\, 0.7) \,\cup\, [0.7,\, 1.3] \,\cup\, (1.3,\, 1.5]$$
$$L_{\text{train}} = \{0.70 + 0.05k \mid k = 0, 1, \ldots, 12\}$$



Evaluation Setup

- **Interpolation:** unseen samples within the training range
- **Extrapolation:** beam geometries outside the training range
- Metrics: Relative L_2 error and pointwise displacement comparison

Training and Convergence

Neural Network Setup

Surrogate Mapping

$$(L, x) \to u_z$$

Architecture

$$2 \to 64 \to 64 \to 64 \to 1$$

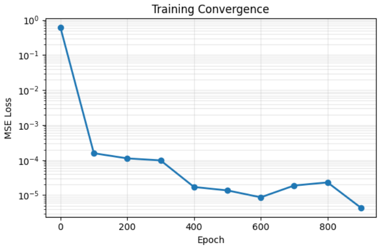
Activation: ReLU

Training

- Optimizer: Adam (10^{-3})
- Batch size: 64
- Epochs: 1000
- Standardized inputs/outputs

Training Loss

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \left(u_{z,i}^{\mathrm{FEM}} - u_{z,i}^{\mathrm{NN}} \right)^2$$



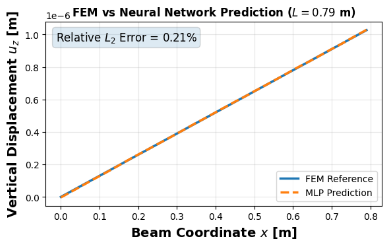
Performance Summary

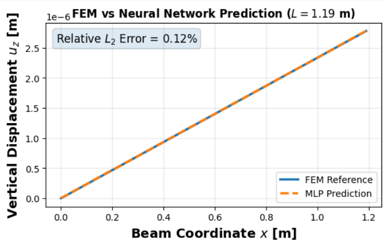
Interpolation L2 Error: 1.1×10^{-3}

Final training Loss: 4.4×10^{-6}

Model Validation: Interpolation & Extrapolation

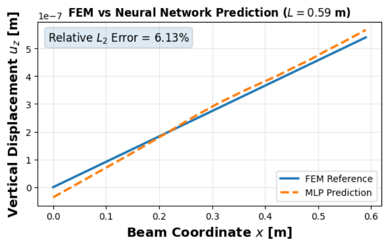
Interpolation Performance

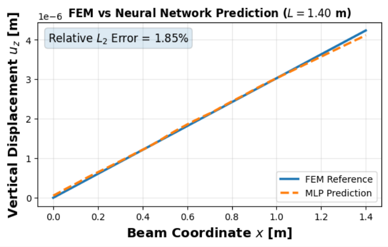




$L \in [0.7, 1.3]$ (unseen samples)

Extrapolation Performance

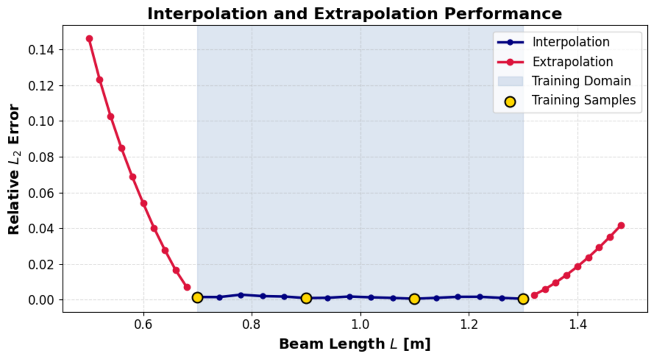




$L < 0.7$ and $L > 1.3$

Error Analysis Across Geometry Space

Relative L_2 Error



Error variation across beam length L

Key Observation

- Low error inside training region
- Increased error in extrapolation regime

Use of Large Language Models (LLMs)

- Large Language Models were used as a supportive tool during development.
- Their usage was limited to:
  - debugging implementation issues (FEniCSx, Python, PyVista),
  - improving visualization and plotting workflows,
  - assisting with LaTeX Beamer formatting.

Conclusion

Overview

- FEM-based dataset with physics consistency
- Geometry-aware sampling across beam configurations
- Centerline representation for compact displacement modeling
- Neural surrogate learned geometry-to-response mapping

Key Results

- High accuracy in interpolation regime
- Physically consistent displacement predictions
- Reduced generalization under extrapolation

Contribution

- End-to-end FEM-to-ML surrogate workflow
- Fast prediction of structural deformation
- Demonstrates simulation-driven ML for design tasks

Resources

Code & Implementation

- **GitHub Repository:**
  https://github.com/adnan-math/beam_fem_ml
- **Colab Notebook:**
  https://colab.research.google.com/github/adnan-math/beam_fem_ml/blob/main/analysis.ipynb

All experiments are fully reproducible and documented